

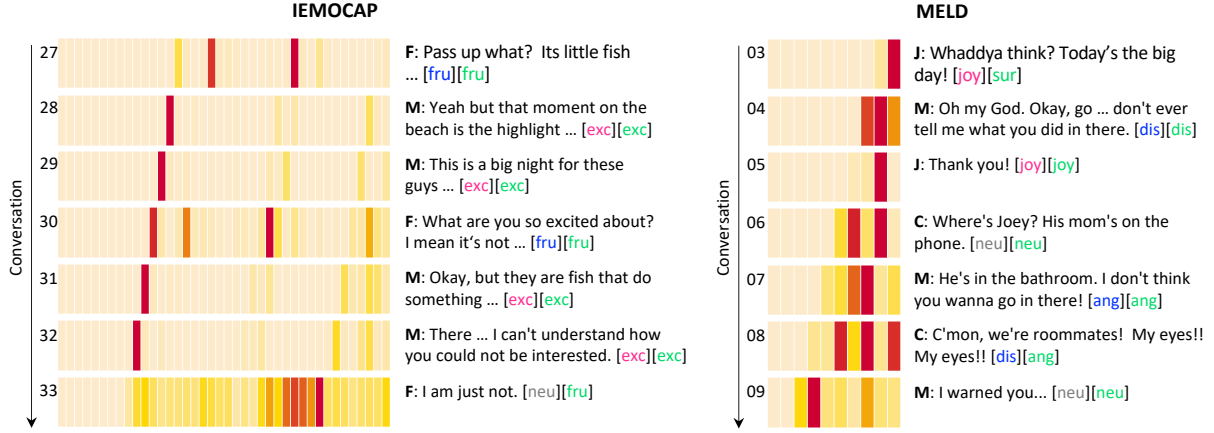


Figure 4: Evolution of memory selection as a conversation develops. The attention weights come from BiF-AGRU for both IEMOCAP and MELD. Each utterance is tagged with two labels, the first is ground truth and the second is the prediction by BiF-AGRU. IEMOCAP: M: Male, F: Female; MELD: J: Joey, M: Monica, C: Chandler.

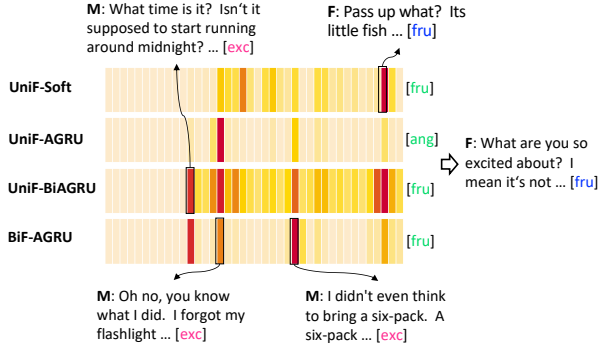


Figure 5: Comparison of memory selection between UniF-Soft and UniF-AGRU, UniF-BiAGRU, and BiF-AGRU.

intermediate ones (see Fig 5) that make her *frustrated*. As for MELD, Joey is joyful all the time and his attention is paid to his last *joyful* utterance. Monica pays her most attention to the conversation between her and Joey, providing clues for Chandler. Chandler focuses on Joey and where Joey is.

Attention Comparison. The selection of memory also varies between different attention mechanisms. In Fig 5, given a query utterance expressed by the Female with *frustrated* emotion, Soft Attention focuses on the utterance that also expresses *frustrated*. AGRU pays most attention to one utterance that could be the reason for the Female’s *frustrated* emotion, but it classifies the query emotion as *angry* in this example. BiAGRU can sense both kinds of clues, providing more comprehensive memory. BiF improves the memory representation and helps AGRU to extract the memory as comprehensively as UniF-BiAGRU.

Error Analysis. In Fig 4, the 33rd utterance of IEMOCAP is recognized as *frustrated* not *neutral*. We argue that the original annotation might not be accurate. Given several turns of

frustrated and the latest response (34th) from the Male, the Female could express *frustrated* again. Still, we cannot deal with minority classes very well on MELD, such as *disgust*. With more data or multimodal features to disambiguate with other emotions, this issue might be better resolved.

Conclusions

We propose an Attention Gated Hierarchical Memory Network (AGHMN) for Real-Time Emotion Recognition. Firstly, the proposed Hierarchical Memory Network improves the quality of utterance features and memories. Then, the proposed Attention GRU summarizes better contextual information than the commonly used Soft Attention. We conduct extensive experiments on two emotion conversation datasets, and our models outperform the state-of-the-art approaches with significant margins. Lastly, ablation studies and attention visualization demonstrate the efficacy of each component of our AGHMN models.

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